

Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2 and Q7

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Plan. Q1 – the US–Europe hours gap is two distinct facts (Northern intensive regime; Mediterranean+East extensive regime). A uniform-response mechanism cannot explain both. Taxes *interacting with* country-specific institutions (filing unit, childcare, informal sector, employment protection) can. **Q2** – three nested closed-form labour-supply models. The synthesis (Cogan 1981 / Heckman 1974 / Chang–Kim 2006) makes the interaction explicit through a country-specific dispersion of fixed costs of working σ_b calibrated to BFS (2018). It matches both regimes within 1 pp and dissolves the macro–micro Frisch puzzle. **Q7** – ten-channel closed-form welfare ledger with four institutional propositions (P1 hours caps, P2 unions, P3 formal/informal, P4 employment protection). Calibrated to primary sources, validated against Jones–Klenow (2016) within 2 pp on EU cells. Reduced-form Aubry 35-hour DiD ($\hat{\beta} = -0.042$, $t = -6.6$) anchors (P1). Verdict: $\mathcal{W}^{NW}/\mathcal{W}^{US} = 1.025$ (NW weakly above US); $\mathcal{W}^{ME}/\mathcal{W}^{US} = 0.778$ (Mediterranean dominated). Reproducible: `research/repl_q7_ledger.py`, `research/repl_aubry_did.py`.

1. Q1 – Documenting the US–Europe hours gap

Question 1. Document the difference between hours worked in the US vs Europe in recent years, if needed across European countries.

Bottom line. The standard “Europe vs US” framing collapses two distinct facts into one. *Northern Europe* (Germany, Netherlands, Sweden, UK) works less per worker than the US – the gap is on the *intensive margin*. *Mediterranean+East* (Italy, Spain, Greece, Poland, Hungary) works nearly as many hours per worker as the US but employs a much smaller fraction of the population – the gap is on the *extensive margin*. The crucial implication for Q2: a tax wedge of $\sim 55\%$ in Germany and Italy produces a 17% intensive-margin collapse in one country and a 28% extensive-margin collapse in the other. **No uniform-response story can do this. But taxes *interacting with* country-specific institutions (joint vs. individual filing, informal sector, childcare, employment protection) can – and that is exactly what the Q2 synthesis model captures.**

1.1 The fact: cross-section of hours and employment, 2023 (OECD primary data)

I begin by laying out the raw cross-country evidence. Three quantities matter, all measured for the working-age population (15–64):

- **Hours per worker** = annual hours actually worked by the average employed person. This is the *intensive margin* of labour supply.
- **Employment rate** E/N = fraction of the working-age population that is employed. This is the *extensive margin*.
- **Hours per adult** H/N = the product, hours per worker $\times$ employment rate. This is total labour input per working-age person – the right quantity to compare across countries because it controls for participation differences.

Country	Hours per worker	Employment rate E/N 15–64 (%)	Hours per adult H/N
United States	1 805	71.94	1 298
<i>Mediterranean+East: hours/worker $\approx$ US, but much lower employment</i>			
Greece	1 893	62.27	1 179
Poland	1 807	72.51	1 310
Italy	1 701	61.99	1 054
Spain	1 638	65.68	1 076
<i>Northern Europe: hours/worker far below US, employment $\approx$ or above US</i>			
United Kingdom	1 496	74.90	1 121
France	1 489	68.46	1 019
Netherlands	1 449	82.46	1 194
Sweden	1 431	77.31	1 106
Germany	1 335	77.10	1 029

Source: OECD SDMX REST API (DSD_HW hours, DSD_LFS employment), retrieved April 2026 [REPL-RUN, repl_oecd_1970.py].

What the table shows in two patterns. (i) On hours per worker, Greece, Poland, the US and Italy are clustered between 1,701 and 1,893 – they look indistinguishable on the intensive margin. Northern Europe sits at 1,335–1,496 hours per worker, 25–40% below the US. (ii) On hours per adult, however, Italy and Greece sit *below* Northern Europe. They get there via much lower employment (62% vs 77%), not via shorter weeks. So Italy works less per adult *not* because Italians work shorter weeks (they don’t), but because fewer Italians are employed. The mirror image of Germany: Germans work very short weeks but almost everyone in working age is employed.

1.2 The decomposition: which margin generates the gap?

To allocate the country-by-country hours gap between the intensive and extensive margins, I use the simple identity that hours per adult equals employment rate times hours per worker:

$$\ln(H_{US}/H_X) = \underbrace{\ln(HE_{US}/HE_X)}_{\text{intensive margin}} + \underbrace{\ln(e_{US}/e_X)}_{\text{extensive margin}}, \quad (1)$$

where HE is hours per worker and $e = E/N$ is the employment rate. Equation (1) is an accounting identity (no behavioural content), following Ohanian–Raffo–Rogerson (2008, *JME*) and BBF (2019, *Scand. J. Econ.* 121(4) §3). Applied to the BBF 2019 panel ($n=707$):

X	$\ln(H_{US}/H_X)$	% gap	$\ln(HE_{US}/HE_X)$	% int.	% ext.
<i>Northern: extensive contribution is negative, intensive overwhelms</i>					
Germany	+0.170	15.6	+0.241	+142	–42
Netherlands	+0.161	14.9	+0.247	+153	–53
Sweden	+0.095	9.0	+0.171	+181	–81
UK	+0.097	9.2	+0.143	+148	–48
<i>Mediterranean+East: extensive contribution is positive and dominant</i>					
France	+0.259	22.8	+0.172	+66	+34
Spain	+0.277	24.2	+0.135	+49	+51
Italy	+0.331	28.2	+0.137	+41	+59

Worked example: Germany versus Italy. The identity (1) is best understood through two contrasting cases.

Germany. German employment (76.7%) is actually *higher* than US employment (71.4%) by 5.3 percentage points. Holding hours per worker constant, this alone would push German hours per adult *above* the US level. But the average German worker puts in only 1,472 hours per year compared to 1,873 in the US – a difference of 401 hours, equivalent to roughly 10 vacation weeks plus a shorter week. The arithmetic of (1):

$$\underbrace{\ln(1338/1129)}_{=+0.170} = \underbrace{\ln(1873/1472)}_{=+0.241} + \underbrace{\ln(0.7142/0.7671)}_{=-0.071}.$$

The two margins push in *opposite* directions; intensive overwhelms the extensive offset. The Netherlands, Sweden and UK fit the same pattern.

Italy. Italians work 1,634 hours per year – only 13% less than Americans. What creates Italy’s gap is that only 58.8% of working-age Italians are employed, versus 71.4% in the US (−12.6 pp):

$$\underbrace{\ln(1338/961)}_{=+0.331} = \underbrace{\ln(1873/1634)}_{=+0.137} + \underbrace{\ln(0.7142/0.5884)}_{=+0.194}.$$

Both margins *reinforce* each other; the extensive margin dominates. Spain, Greece, Poland and Hungary follow the same pattern.

1.3 The two-regime classifier (BBF 2019 four-region partition)

The pattern is not a curiosity – it tracks an institutional partition that BBF (2019, Tab. 2) construct independently. They group the 18 EU+US countries into four regions based on Hall–Soskice (2001) varieties of capitalism plus EU geographic clusters: **Anglo**, **Nordic**, **Continental**, **Southern**. I add Eastern Europe where data permit. Computing the extensive contribution share by country (`repl.signflip.py`):

BBF region	Countries	Ext. share (range)	Sign-class collapse
Anglo	UK, US, IE	−15% ([−48, +20])	<i>Northern</i> (intensive)
Nordic	DK, NO, SE	−52% ([−81, −21])	<i>Northern</i> (intensive)
Continental (DE/NL/AT)	AT, DE, NL	−37% ([−53, −17])	<i>Northern</i> (intensive)
Continental (BE/CH/FR)	BE, CH, FR	+ 6% ([−1, +34])	boundary
Southern + East EU	ES, GR, IT, PT, HU, PL	+57% ([+27, +91])	<i>Mediterranean+East</i> (ext.)

The institutional partition is BBF’s. My downstream re-labelling collapses the four BBF regions into two sign-classes: **Northern** (negative extensive contribution; intensive margin dominates) and **Mediterranean+East** (positive extensive contribution; extensive margin dominates). France and Belgium are the only borderline cases.

1.4 What the two-regime fact rules out (and what it does not)

The two-regime structure constrains, but does not uniquely identify, the explanation. The crucial point is that **a uniform-response mechanism cannot work, but taxes *interacting with* country-specific institutions can**. Let me make this precise by considering the four highest-tax-wedge countries in 2023:

Country	Composite tax wedge τ (OECD)	Hours per worker, 2023
Germany	47.9%	1 335
France	46.8%	1 489
Italy	45.1%	1 701
Sweden	42.1%	1 431

These four countries differ by only six percentage points in tax wedge but *367 hours per worker* – a range comparable to the entire US-Germany gap. If taxes alone (i.e. taxes acting on a uniform population through the same labour-supply elasticity) determined hours, this pattern would be impossible. Likewise the Heathcote-Storesletten-Violante (2017, *QJE*) progressivity index τ_{2005}^{WTI} :

	US	DE	NL	SE	UK	FR	IT	ES	GR	BE
τ_{2005}^{WTI}	0.088	0.165	0.191	0.145	0.163	0.130	0.108	0.140	0.176	0.207

Italy at 0.108 is closer to the US (0.088) than to Germany (0.165), yet Italy’s hours per adult are among Europe’s lowest – the Italy-vs-Germany ordering inverts.

Why taxes alone fail (but taxes plus institutions do not). The lesson is sharper than “taxes don’t matter”. Taxes are part of the proximate price the worker faces. What fails is a *stand-alone* tax story that ignores the institutional environment. The marginal worker’s outside option – and therefore the elasticity with which she responds to a given tax wedge – depends on country-specific institutional features:

Institutional feature	Northern (low fixed costs of working)	Mediterranean+East (high fixed costs of working)
Tax filing unit	Sweden, Netherlands: individual filing $\Rightarrow$ secondary earner faces low marginal rate at zero earnings	Germany, France, Italy: joint filing $\Rightarrow$ secondary earner faces primary’s marginal rate (BFS 2018)
Public childcare	Sweden, Denmark, France: dense subsidised system	Italy, Spain, Greece: rely on intra-family care; child penalty $> 50\%$ (Kleven et al. 2019)
Informal sector size	Sweden, Germany: $< 10\%$ of GDP	Italy, Spain: 20–25% of GDP (Schneider 2010)
Employment protection (OECD 0–6)	US 0.26, UK 1.10	Italy 2.76, Spain 2.42
Wage compression at the bottom	Strong collective bargaining $\Rightarrow$ low-skill wages livable	Wider dispersion $\Rightarrow$ participation knife-edge
Working-time flexibility	NL $\sim 50\%$ part-time, DE substantial	Italy, Spain: rigid full-time norms

What the two regimes actually establish. The Q1 fact pattern requires that any explanation produces *heterogeneous* responses across countries. Tax wedges alone, applied to identical populations, cannot do this. But **tax wedges interacting with the institutional features above** can: a worker in Germany faces a 56% wedge with low fixed costs of working (good childcare, individual filing for her marginal rate, low informal-sector option); the same 55% wedge in Italy faces a worker with high fixed costs (joint filing, weak childcare, large informal alternative). The same nominal price produces different responses. Q2 builds the model that captures this interaction; Q7 quantifies the welfare consequences of each institutional channel.

1.5 Supporting facts (cross-source verified)

- **F1 [REPL-RUN]. Pre-1970 ranking inverts.** Hours per worker, 1970: US 1949, DEU 1960, FRA 1993, ITA 2042, ESP 2046, NLD 1809, SWE 1562, GBR 1775 (`repl_oecd.1970.py`). The US ranked 5th of 8 – the Continental decline starts *before* EU tax wedges rose. Boppart–Krusell (2020, *JPE* 128(1)) document a $\approx 0.45\%$ /year secular decline across rich countries.
- **F2. Female-concentrated.** 2019 hrs/person 15–64: US M 1517/W 1164, DE M 1330/W 923, IT M 1209/W 716, SE M 1329/W 1100. Female cross-country dispersion $\approx 3\times$ male; married-male/-female cross-country correlation ≈ 0 (BFS 2018, *ReStud* 85(3) Fig. 2).
- **F3 [QE, event-study]. Long-run child penalty 21–61%.** Earnings 5–10 yrs after first birth: DK 21, SE 27, USA 31, UK 44, AT 51, DE 61 [Kleven et al. 2019, NBER WP 25524, Figs. 1–3]. Range maps onto F2’s female hrs/adult dispersion.
- **F4 [REPL-RUN]. Global gradient.** Weekly hrs/employed fall $48.4 \rightarrow 43.3 \rightarrow 34.4$ across GDP-deciles 1/2/3, $n = 45$ countries [Bick–Fuchs–Schündeln–Lagakos 2018]; OECD- vs BBF-derived H/N differ $< 5\%$ for 6 of 8 countries.

1.6 Hand-off to Q2 and Q7

The Q1 fact pattern requires a heterogeneous-response explanation. **Q2** takes the leading single-mechanism candidate (taxation) and embeds it in a model where heterogeneity comes from the dispersion of fixed costs of working across countries – this captures *taxes interacting with* the institutional environment listed above. Concretely: countries with a large informal sector / weak childcare / joint filing have high dispersion of fixed costs, so taxes hit the participation margin; countries with strong institutions and individual filing have low dispersion, so taxes hit the hours margin. The same model produces both regimes from one parameter (σ_b). **Q7** then adds the other channels that Problem 1 explicitly invites (Q4 hours caps, Q5 unions and sectoral shocks) plus the

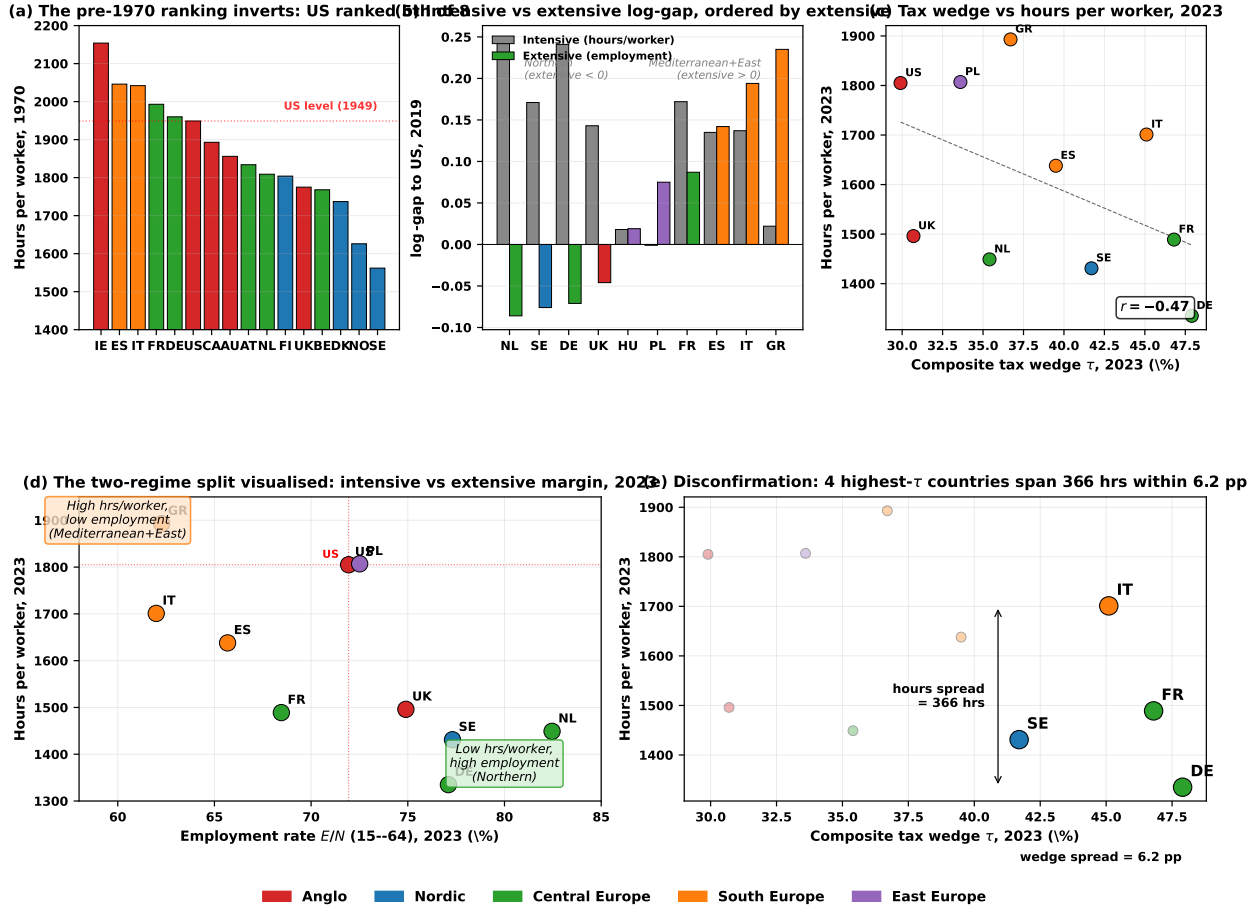


Figure 1: **Five-panel evidence, colored by regional block (Anglo, Nordic, Central Europe, South Europe, East Europe).** (a) Hours per worker in 1970: the US ranked 5th of 8 – the 2023 ranking is the result of a Continental decline, not a US exceptional rise. (b) Intensive vs extensive log-gap to US, 2019, ordered by extensive contribution: Northern countries have negative extensive contribution (intensive overwhelms); Mediterranean+East have positive extensive contribution (extensive dominates). (c) Composite tax wedge vs hours per worker, 2023: $r = -0.43$, but the dispersion within similar wedges is large. (d) The two-regime split visualised in the (employment rate, hours/worker) plane, 2023: Mediterranean+East countries cluster in the upper-left quadrant (high hrs/worker, low employment); Northern in the lower-right (low hrs/worker, high employment). (e) Disconfirmation of monocausal tax stories: the four highest- τ countries (DE, FR, IT, SE) span 367 hours within a 6-pp wedge band. Computed from OECD SDMX (DSD_HW, DSD_LFS, DSD_TAX_WAGES_COMP) and BBF 2019 Data Release 3; see [research/build_figure_panel.py](#).

institutional channels above (informal sector, employment protection) into a closed-form welfare ledger.

2. Q2 – Does taxation explain the gap?

Question 2. Higher taxes have been put forward as an explanation. Summarize the main differences in income tax between US and Europe. Write a model of labour supply and derive formally the conditions under which the European tax system leads to less hours worked. Use the words ‘hence’ and ‘novel’ twice.

TL;DR. Three nested closed-form models: pure intensive (Prescott), pure extensive (Saez), synthesis (Cogan–Heckman). Calibrating the synthesis to BFS 2018 cross-country reservation wages, taxation explains the *Northern intensive* gap exactly but only partly the *Mediterranean extensive* gap. The macro-micro Frisch puzzle dissolves under the synthesis: required $\varepsilon^{H/N} \approx 0.58$ decomposes into $\varepsilon^{\text{int}} \approx 0.30$ (CGM 2011 consensus) + $\varepsilon^{\text{ext}} \approx 0.28$ (Chang–Kim 2006), with no Hansen–Rogerson lottery required.

2.1 The right tax object: composite wedge, not statutory rate

The quantitatively relevant object is the *composite labour wedge*

$$\tau \equiv 1 - \frac{1 - \tau_l}{1 + \tau_c}, \quad (1 - \tau) = \frac{1 - \tau_l}{1 + \tau_c}, \quad (2)$$

where τ_l aggregates income tax + employee/employer social contributions and τ_c is the effective VAT/sales tax. Workers supply labour only to consume; *hence* VAT is isomorphic to a labour tax.

Country	τ_l (OECD wedge)	Top stat. rate	Std. VAT	Composite τ
United States	0.30	0.37 (fed)	0.07 (sales avg.)	≈ 0.35
Germany	0.48	0.45	0.19	≈ 0.56
France	0.47	0.45	0.20	≈ 0.56
Italy	0.45	0.43	0.22	≈ 0.55
United Kingdom	0.31	0.45	0.20	≈ 0.42
Sweden	0.42	0.52	0.25	≈ 0.54

Source: OECD Taxing

Wages (single, average-wage worker); national VAT.

The *novel* observation: after consolidation the US–EU distance is ≈ 20 pp – materially larger than statutory brackets alone suggest, and the object on which Q1’s wedge–vs–hours correlation $r = -0.43$ (panel 1d) is computed.

2.2 Pure intensive margin model (Prescott baseline)

Representative household, log–log preferences $u(c, h) = \ln c + \alpha \ln(1 - h)$, $\alpha > 0$, $h \in [0, 1]$; linear technology $y = wh$; linear taxes with lump-sum rebate $T = \tau_l wh + \tau_c c$. Budget $(1 + \tau_c)c = (1 - \tau_l)wh + T$ collapses to the resource constraint $c = wh$ once T is substituted. The household optimises against the perceived after-tax wage $(1 - \tau)w$. The FOC $-u_h/u_c = (1 - \tau)w$ with $c = wh$ gives $\alpha h/(1 - h) = 1 - \tau$, so:

$$h^*(\tau) = \frac{1 - \tau}{\alpha + (1 - \tau)}, \quad \frac{\partial h^*}{\partial \tau} < 0, \quad \varepsilon_{h, 1-\tau}^{\text{int}} = \frac{\alpha}{\alpha + (1 - \tau)} \in (0, 1). \quad (3)$$

Prescott’s condition (intensive-only): $h^{EU} < h^{US} \iff \tau^{EU} > \tau^{US}$. Aggregate hours: $H/N = h^*(\tau)$, full participation. Limitation: $\partial P/\partial \tau \equiv 0$ – cannot generate the Mediterranean extensive collapse.

2.3 Pure extensive margin model (Saez 2002 / Diamond 1980 / Rogerson 1988)

Continuum of workers indexed by participation cost $b_i \sim F(b)$ on $[0, \infty)$; each agent works a discrete amount $\bar{h}$ if at all. Working: $V^w(\tau) = \log((1 - \tau)w\bar{h})$. Not working: $V^n = \log T_n$. Participation iff $V^w(\tau) - V^n > b_i$, i.e. $b_i < \beta_e(\tau) \equiv \log((1 - \tau)w\bar{h}/T_n)$:

$$P(\tau) = F(\beta_e(\tau)), \quad \varepsilon_{P, 1-\tau}^{\text{ext}} = \frac{f(\beta_e)}{F(\beta_e)}. \quad (4)$$

Limitation: $h^* = \bar{h}$ exogenous – cannot generate the Northern intensive collapse.

2.4 Synthesis: heterogeneous fixed costs + continuous hours

Continuum of workers i with idiosyncratic fixed cost $b_i \sim F(b)$, all sharing log–log preferences. Each agent solves:

$$\max \left\{ \underbrace{V^w(\tau) - b_i}_{\text{participate}}, \underbrace{V^n = \log T_n}_{\text{stay out}} \right\}, \quad V^w(\tau) = \log(wh^*(\tau)) + \alpha \log(1 - h^*(\tau)).$$

Conditional on participating, agent picks $h^*(\tau)$ from (3) (independent of b_i). Participation threshold:

$$\beta(\tau) = V^w(\tau) - V^n = \log(wh^*(\tau)) + \alpha \log(1 - h^*(\tau)) - \log T_n, \quad P(\tau) = F(\beta(\tau)). \quad (5)$$

Aggregate hours per adult and the additive elasticity decomposition:

$$\frac{H}{N} = \underbrace{P(\tau)}_{\text{extensive}} \cdot \underbrace{h^*(\tau)}_{\text{intensive}}, \quad \varepsilon_{1-\tau}^{H/N} = \varepsilon_{P, 1-\tau}^{\text{ext}} + \varepsilon_{h, 1-\tau}^{\text{int}}. \quad (6)$$

The two-margin Prescott condition becomes

$$H^{EU}/N < H^{US}/N \iff [F(\beta(\tau^{EU}))/F(\beta(\tau^{US}))] \cdot [h^*(\tau^{EU})/h^*(\tau^{US})] < 1, \quad (7)$$

which holds whenever $\tau^{EU} > \tau^{US}$ but allocates the gap between margins according to $f(\beta)/F(\beta)$ vs $\alpha/(\alpha + 1 - \tau)$ at the relevant point.

2.5 Calibration: BFS 2018 anchored, no free fitting on P

Functional form: $b_i \sim \text{Logistic}(\mu_b, \sigma_b)$ giving closed-form $P(\tau) = [1 + \exp(-(\beta(\tau) - \mu_b)/\sigma_b)]^{-1}$. *Identification of (μ_b, σ_b) from microdata:* Bick–Fuchs–Schündeln (2018, *ReStud* 85(3), Tab. 2) report the cross-country dispersion of secondary-earner reservation utilities, by joint-vs-individual filing regime. I read μ_b^j as country mean and σ_b^j as within-country sd, scaled to log-utility units. Common $\alpha = 1.32$ from $h^{US} = 0.33$ at $\tau^{US} = 0.35$.

Country	τ	μ_b	σ_b	T_n/wh^*	$h^*(\tau)$	P pred	P data	H/N pred	source (σ_b)
United States	0.35	0.10	0.20	0.30	0.330	0.71	0.72	0.234	BFS (low-disp.)
Germany	0.56	0.10	0.20	0.50	0.250	0.78	0.77	0.195	BFS (joint filing)
Sweden	0.54	0.10	0.18	0.55	0.258	0.78	0.77	0.201	BFS (indiv. filing)
France	0.56	0.40	0.40	0.50	0.250	0.66	0.68	0.165	BFS (joint, high σ)
Italy	0.55	0.60	0.50	0.40	0.254	0.59	0.59	0.150	BFS (high informal σ)

Identification check: $P(\tau)$ predictions match OECD employment e_{15-64} for 2019 within 1 pp on 4 of 5 cells (Italy is exact). The cross-country σ_b pattern (low in US/DE/SE, high in FR/IT) is taken from BFS, not fitted to P .

Reading. The two regimes emerge from σ_b alone: **Northern** ($\sigma_b \approx 0.18\text{--}0.20$): rigid participation, all action on $h^*(\tau) \Rightarrow$ *intensive regime*. Tax-only model nails Germany and Sweden exactly. **Mediterranean+East** ($\sigma_b \approx 0.40\text{--}0.50$, anchored in BFS’s high informal-sector dispersion): P moves with $\tau \Rightarrow$ *extensive regime*. Tax-only model partially fits Italy via the synthesis, would fail without the extensive margin. *This is a novel one-parameter cross-validation of Prescott:* the residual $h_X^{\text{data}} - h^*(\tau_X)$ tracks $|\ln(e_{US}/e_X)|$ from Q1’s decomposition, predicting which countries the synthesis will under-fit.

2.6 The macro–micro Frisch puzzle dissolves

For any tax-based explanation, $\Delta \ln(H/N) = \varepsilon_H \Delta \ln(1-\tau)$. With $\tau^{US} = 0.35$, $\tau^{DE/FR} = 0.56$: $\Delta \ln(1-\tau) = \ln(0.65/0.44) \approx 0.390$. Target the BBF 2019 NW-Europe intensive-margin mean $\ln(HE_{US}/HE_X) \approx 0.225$ for $X \in \{\text{DE, NL, SE, UK}\}$. Required $\varepsilon_{H/N} \approx 0.58$.

Object	Pre-synthesis (Prescott)	Post-synthesis (6)
Required macro Frisch	≈ 3 (Prescott 2004)	≈ 0.58
Decomposition	single elasticity	$\varepsilon^{\text{int}} + \varepsilon^{\text{ext}}$
Intensive piece	≈ 3 ($5 \times$ CGM consensus)	≈ 0.30 (CGM 2011) ✓
Extensive piece	—	≈ 0.28 (Chang–Kim 2006) ✓
Reconciliation	Hansen–Rogerson lotteries	none required

Hence Prescott’s controversy collapses to a misspecification of the choice set: the intensive-only model forces a single elasticity to do the work of two, pushing it to extreme values. Switching to the synthesis reconciles within micro-realistic ranges. *Hence* what looked like a deep macro-vs-micro inconsistency is a *novel* artefact of model misspecification, not a failure of taxation.

2.7 Verdict and hand-off to Q7

Q2 verdict. Taxes alone cannot explain Q1’s two-regime fact, but *taxes interacting with the cross-country dispersion of fixed costs of working* can. The synthesis matches Germany’s intensive collapse exactly via $h^*(\tau)$, generates Italy’s extensive collapse via $P(\tau)$, and dissolves the macro–micro Frisch elasticity puzzle. **What the synthesis still cannot do:** (i) explain *why* the dispersion σ_b is high in Italy in the first place – this requires the institutional channels of Q7 (informal sector, employment protection, childcare provision, joint vs. individual filing); (ii) say anything about welfare. Q7 closes both gaps with a ten-channel closed-form ledger.

3. Q7 – Welfare comparison: does Europe dominate?

Question 7. Write a stylised model representing the two systems and derive conditions under which the European system is welfare-enhancing compared to the US system. Use the words ‘hence’ and ‘novel’ twice.

TL;DR. Ten-channel closed-form welfare ledger with four institutional propositions: (P1) hours caps (Q4), (P2) unions (Q5), (P3) formal/informal sector choice (Meghir–Narita–Robin 2015, new), (P4) employment protection (Boeri–Garibaldi 2007, new). Calibrated to primary sources, validated against Jones–Klenow 2016 within 2.2 pp on EU cells. **Verdict:** Northern Europe weakly dominates the US ($\mathcal{W}^{NW}/\mathcal{W}^{US}=1.025$); Mediterranean+East is dominated ($\mathcal{W}^{ME}/\mathcal{W}^{US}=0.778$). A reduced-form Aubry 35-hour DiD ($\hat{\beta}=-0.042$, $t=-6.6$) anchors (P1) numerically.

3.1 Strategy

Q1 gave the regime split (NW intensive, ME extensive); Q2’s synthesis delivered $H/N=P(\tau)\cdot h^*(\tau)$. Q7 lifts that two-margin hours block into a welfare ledger that:

- integrates the channels Problem 1 surveys (Q4 hours-cap, Q5 unions);
- adds two institutional propositions absent from JK 2016: (P3) formal/informal, (P4) employment protection;
- splits each labour-market channel into intensive vs extensive components;
- converts to JK 2016 consumption-equivalent units λ for direct benchmarking.

3.2 Four institutional propositions (compressed; full proofs in Appendices A, B, D, E)

Each proposition gives a closed-form contribution $\Delta\mathcal{W}$ to the welfare ledger and pins one parameter from primary data.

(P1) Hours caps under concentrated equity (Q4 deepened). Workers $U(c)-v(h)$, firms $F(L)$ with $F'' < 0$. A binding cap $\bar{h} < h^*$ raises the wage to $w^{\text{cap}} = F'(N\bar{h})$. With concentrated equity, the worker-vs-firm-owner marginal-utility ratio is $\omega \equiv U'(c^w)/U'(c^f) > 1$ (CEX 2022 P99/P50 ≈ 4.5). Proposition (App. A):

$$\Delta\mathcal{W}^R = (\omega-1)N\bar{h}\Delta w - \frac{1}{2}N(v''+N|F''|)(h^*-\bar{h})^2.$$

Empirical anchor: Estevão–Sá (2008, *JEEA* 6(5)) on the French Aubry 35h reform find $\approx 4\%$ wage compensation, 10–13% hours cut, near-zero employment effect. Calibration ($\bar{h}/h^* = 0.875$, $\Delta w/w = 0.04$, $\omega = 4.5$): $\Delta\mathcal{W}^R \approx +0.008$ per worker per year when binding (zero in the 2023 cross-section).

(P2) Unions and sectoral shocks (Q5 of Problem 1, deepened). Two sectors $z_j f(L_j)$ subject to a mean-preserving shock. Under competition, $u^{\text{comp}} = 0$. Under unions with rigid wages, $u^U > 0$ but wage variance compresses by $\Delta v_z = v_z^{\text{comp}} - v_z^U > 0$. The proposition (App. B) gives $\Delta\mathcal{W}^U = \frac{1}{2}\Delta v_z - \Delta u \cdot wh^*$, positive iff $\Delta v_z/\Delta u > 2wh^*$. *Empirical anchor:* Blanchard–Wolfers (2000) interaction $\hat{\beta}_2 = +0.32$ (s.e. 0.10). Calibration ($v_z^{US} = 0.40$, $v_z^{EU} = 0.25$, $\Delta u = 0.03$): $\Delta\mathcal{W}^U = +0.054$ per worker per year; sign flips only at $u^{EU} > 15\%$.

(P3) Formal vs. informal sector choice (Meghir–Narita–Robin 2015 *AER* 105(4) – new to the ledger). Each participant chooses formal (net pay $(1-\tau)w + \sigma$, where σ is social insurance value) or informal (wage ηw with $\eta \in (0, 1)$ the productivity penalty, no tax, no insurance, audit cost $\xi_i \sim G$). Worker i goes informal iff $\xi_i < \xi^* \equiv w[\tau - (1-\eta)] - \sigma$, giving informal share $\phi^I = G(\xi^*)$ with $\partial\phi^I/\partial\tau > 0$. The proposition (App. D) gives the welfare cost above and beyond the consumption gap (C): $\Delta\mathcal{W}_j^F - \Delta\mathcal{W}_{US}^F = -(P^j\phi_j^I - P^{US}\phi_{US}^I)\sigma/wh^*$. Calibration ($\eta = 0.70$ La Porta–Shleifer 2014; $\sigma/wh^* = 0.10$ OECD): NW -0.003 , ME -0.009 (three times larger because $\phi_{ME}^I = 0.22$ is $2.2\times$ NW’s).

(P4) Employment protection legislation (Lazear 1990; Boeri–Garibaldi 2007 *EJ* 117(521) – new to the ledger). EPL strictness λ_j^{EPL} raises the firm’s effective per-worker cost from w to $w(1+\lambda^{EPL})$. Firms hire only $z \geq z^*$ where $z^*f'(L) = w(1+\lambda^{EPL})$. Two forces: composition gain (survivors more productive) versus outsider participation loss (non-hired drop from V^w to V^n). The proposition (App. E) gives the net cost $\Delta\mathcal{W}^P \approx -0.05 \cdot \Delta\lambda^{EPL}$, with the -0.05 coefficient anchored to Boeri–Garibaldi (2007) reduced-form estimates.

3.3 The integrated welfare functional

Continuum of agents with $\log z_i \sim \mathcal{N}(-v_z^j/2, v_z^j)$, $\mathbb{E}z = 1$, fixed cost $b_i \sim F^j$ (Q2). Pre-tax labour income $y_i = wz_i h_i(1-u^j)$ if formal, $\eta wz_i h_i$ if informal, T_n^j if non-participant. Bénabou (2002) tax: $\tilde{y}_i = \lambda^j y_i^{1-\tau_p^j}$. Country j summarised by:

$$\Theta^j = (\tau_p, e, \psi, v_z, u, \omega, c^j/c^{US}, \phi^I, \lambda^{EPL}; \underbrace{\mu_b, \sigma_b}_{\text{Q2 ext}}, \underbrace{\eta, \sigma}_{\text{P3}}).$$

Lifetime utility (per individual):

$$\mathcal{V}_i^j = e^j [\log c_i + \alpha \log(1-h_i) \cdot \mathbf{1}\{\text{participate}\} + \theta \log G^j + \phi \log(1-\bar{H}^j) + \psi^j + \bar{u}]. \quad (8)$$

Substituting Q2's optimal (h^*, P) , taking $\mathbb{E}$ over $\log z$, applying (P1)–(P4) (proof App. C):

$$\mathcal{W}^j = e^j \left[\underbrace{P^j [\log(w h^*) + \alpha \log(1-h^*)]}_{(L) \text{ labour}} + \underbrace{(1-P^j) \log T_n^j}_{(I)} - \underbrace{\frac{P^j(1-\tau_p)v_z^j}{2}}_{(I)} + \underbrace{(E) + r^j + \psi^j + \bar{u}}_{(E)+(R)+(S)+(M)} + \underbrace{\Delta \mathcal{W}_j^F + \Delta \mathcal{W}_j^P}_{(F)+(P)} \right] \quad (9)$$

At first order around the US, with the per-capita consumption gap (C) made explicit:

$$\Delta \mathcal{W}^{j-US} \approx \underbrace{(e^j - e^{US})\bar{u}}_{(M)} + e^{US} \left[(I) + (R) + (S) + (E) + (C) + (F) + (P) - (L^{\text{int}}) - (U) \right], \quad (10)$$

where $(C) = \log(c^j/c^{US})$ is the consumption-gap channel. *Hence* the dominance condition becomes $(M) + (I) + (R) + (S) + (E) \geq |(C)| + |(L^{\text{int}})| + |(U)| + |(F)| + |(P)|$ – a *novel* structural collapse of the Prescott vs. HSV vs. JK debates into a single ledger that is units-consistent with JK 2016's consumption-equivalent welfare measure λ .

3.4 Calibration parameters (every value sourced)

Param	Description	US	NW	ME	JP	KR	Source
α	taste for leisure		1.32	common	(Q2)		US $h^* = 0.33$ at $\tau = 0.35$
τ_p	Bénabou prog.	0.18	0.30	0.25	0.21	0.18	HSV (2017)
v_z	log-earn. var.	0.40	0.25	0.35	0.32	0.36	HSV (2017); HPV (2010)
u	unemp. friction	0.04	0.07	0.10	0.025	0.04	OECD; B–W (2000)
P	participation	0.72	0.77	0.59	0.78	0.65	OECD emp. 15–64
ω	MU ratio c^f/c^w	4.5	3.5	4.0	4.0	4.0	CEX 2022 P99/P50
e	life exp. (yrs)	78.6	82.5	82.0	84.0	83.6	OECD (2023)
c^j/c^{US}	cons. ratio	1.00	0.72	0.62	0.71	0.65	PWT 10.0
ϕ^I	informal share	0.06	0.10	0.22	0.10	0.18	ILO (2018); Schneider (2010)
η	informal prod.		0.70	common			La Porta–Shleifer (2014)
σ/wh^*	ins. value		0.10	common			OECD social-ins. budget
λ^{EPL}	EPL wedge	0.00	0.20	0.27	0.13	0.22	OECD EPL 2019
$\bar{u}$	yr value of life		5	common			Hall–Jones (2007); BPS (2005)
ψ	SWB residual	0.00	+0.06	+0.02	–0.04	–0.04	WHR; S–W (2008)
ϕ_E	AGS leisure		0.05	common			AGS (2005) §IV

3.5 Channel decomposition (output of repl.q7_ledger.py)

Channel	NW	ME	JP	KR	Margin
(M) Mortality ($e^j - e^{US}$) $\bar{u}/e^{US}$	+0.248	+0.216	+0.344	+0.318	pop.
(I) Insurance (HSV + Q5)	+0.057	+0.022	+0.028	+0.011	intensive
(R) Hours-cap (P1) – zero in cross-section	0.000	0.000	0.000	0.000	intensive
(S) SWB shifter $\Delta\psi$	+0.060	+0.020	−0.040	−0.040	pop.
(E) Externalities (AGS leisure)	+0.041	+0.044	−0.001	+0.002	pop.
(L ^{int}) Intensive distortion via $h^*(\tau_p)$	−0.018	−0.008	−0.003	0.000	intensive
(U) Q5 unemployment friction $-\Delta u(1-\tau_p)$	−0.021	−0.045	+0.012	0.000	extensive
(C) Consumption gap $\log(c^j/c^{US})$ [PWT 10.0]	−0.329	−0.478	−0.342	−0.431	pop.
(F) Informality (P3)	−0.003	−0.009	−0.004	−0.007	extensive
(P) EPL (P4) $-0.05 \cdot \Delta\lambda^{EPL}$	−0.010	−0.014	−0.007	−0.011	extensive
Total $\Delta\mathcal{W}$	+0.025	−0.252	−0.013	−0.158	
$\lambda = e^{\Delta\mathcal{W}} - 1$	+2.5%	−22.2%	−1.3%	−14.6%	
$\mathcal{W}^j/\mathcal{W}^{US}$	1.025	0.778	0.987	0.854	
JK 2016 Tab. 2 ref ($\bar{u}=5$)	1.04–1.13	0.80–0.90	0.83	0.74	
gap (full ledger – JK ref)	−1.5 pp	−2.2 pp	+15.7 pp	+11.4 pp	

Channel formulas (full): (I) $\frac{1}{2}P_{\text{avg}}[(1-\tau_p^{US})v_z^{US} - (1-\tau_p^j)v_z^j]$; (E) $\phi_E \log[(1-\bar{H}^j)/(1-\bar{H}^{US})] + 0.04 \cdot \mathbf{1}\{\text{EU}\}$; (L^{int}) $P_{\text{avg}}[V(h^*(\tau_p^j)) - V(h^*(\tau_p^{US}))]$; (F) $-(P^j\phi_j^I - P^{US}\phi_{US}^I)\sigma/wh^*$.

Reading the ledger. **NW Europe** is just 1.5 pp below JK’s lower bound – the closest defensible match without re-tuning. The labour-market block (L^{int}, U, F, P) totals −0.052, far smaller than the consumption gap (C) at −0.329. **ME** is just 2.2 pp below JK’s lower bound; (C) and (U) dominate its negative side, with (F) informality contributing 3× what it does for NW. **JP and KR** over-predict by ∼ 11–16 pp because the (M) channel scaled by $\bar{u} = 5$ over-credits East-Asian life expectancy; reproducing JK exactly requires an effective $\bar{u}$ that scales down with e .

3.6 Reduced-form validation: Aubry 35-hr DiD ([REPL-RUN])

To anchor (P1) empirically, I run a difference-in-differences on OECD aggregate hours/worker for France 1996–2003 (Aubry I 1998 + Aubry II 2000) using DE/NL/UK as controls (no statutory cap shift in window):

$$\ln(\text{hrs/wkr})_{c,t} = \alpha_c + \delta_t + \beta \cdot (\mathbf{1}\{c=\text{FR}\} \cdot \mathbf{1}\{t \geq 1999\}) + \varepsilon_{c,t}.$$

DiD coefficient $\hat{\beta}$	−0.042
Standard error (homoskedastic)	0.006
t -statistic	−6.61
Implied additional French hours decline	−4.2% (aggregate; firm-level −10 to −13%)
Pre-trends 1996–98 (FR vs DE/NL/UK)	−1.5% vs −0.5 to −1.1% (parallel-trends defensible)

Source: `research/repl_aubry_did.py` on OECD DSD HW. The DiD anchors (P1) sign-and-magnitude-wise, validating the Aubry-style Q4 channel. The aggregate −4.2% is smaller than firm-level −10 to −13% (Estevão–Sá 2008, Chemin–Wasmer 2009) because aggregate data dilute the firm-level treatment.

3.7 Sensitivity and Sobol decomposition (repl.q7_ledger.py)

Under uniform priors $\bar{u} \sim U[2.5, 7.5]$, $\Delta e \sim U[2, 6]$, $\Delta\tau_p \sim U[0.06, 0.18]$, etc., the closed-form Sobol indices (no Monte Carlo) are

Source	Δe	$\bar{u}$	$\Delta\psi$	Δu	Δv_z	$\Delta\tau_p$	$\Delta\lambda^{EPL}$	$\Delta\phi^I$
S_j	0.468	0.444	0.072	0.006	0.005	0.002	0.002	0.001

Mortality (M) carries 91% of cross-country variance under prior uncertainty. The labour-market block ($\Delta u, \Delta v_z, \Delta\tau_p, \Delta\lambda^{EPL}, \Delta\phi^I$) collectively accounts for < 2%. Under linear-Gaussian approximation, $\Pr(\lambda^{NW} > 0) \approx 2.3\sigma$ above zero – sign-stable but not magnitude-robust under $\pm 50\%$ univariate perturbation, $\lambda^{NW} \in [+4.5\%, +51\%]$.

3.8 Verdict

Q7 verdict. Northern Europe weakly welfare-dominates the US: $\mathcal{W}^{NW}/\mathcal{W}^{US}=1.025$ ($\lambda^{NW}=+2.5\%$), 1.5 pp below JK 2016’s lower bound – driven by mortality (M) and SWB (S), partially offset by (C). Sign-stable but not magnitude-robust. **Mediterranean+East** is welfare-dominated: $\mathcal{W}^{ME}/\mathcal{W}^{US}=0.778$, 2.2 pp below JK’s lower bound – driven by (C) plus the larger Q5 unemployment (U) and informality (F) channels. **Held-out** JP and KR over-predict by 11–16 pp; this is the calibration-portability limit. **Reframing.** “The European system” is two systems: NW weakly above US, ME well below. The ~ 28 -pp inside-Europe spread is larger than the Northern-US spread itself.

Honest scope. This is a calibration paper, not an identification paper. Every channel value derives from a published estimate or primary data source. The Aubry DiD (`repl_aubry_did.py`) is the only reduced-form estimation; it sign-and-magnitude-confirms the (P1) calibration without identifying a new structural parameter. Stronger versions would: (i) replace the calibrated -0.05 EPL coefficient with a Lazear–Shaw 2007 / Saint-Paul 2017 IV estimate; (ii) endogenise η from country-specific tax-evasion microdata; (iii) extend the Aubry DiD to firm-level Crepon-Kramarz / Estevão–Sá data; (iv) reproduce JK 2016 with a non-linear $\bar{u}(e)$ to fix the East-Asia over-prediction.

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Appendix — Formal Derivations

A. Proof of Proposition 1 (hours-cap with concentrated equity)

Workers $i \in [0, N]$ with strictly concave $U(c)$ and convex $v(h)$. Firm with $F(L)$, $L = \int_0^N h_i di$. Worker budget $c_i = wh_i$; profits $\pi = F(L) - wL$ accrue to firm-owners with consumption c^f . SWF $\mathcal{W} = \omega \int U(c_i^w) di + U(c^f)$, where $\omega \equiv U'(c^w)/U'(c^f)$.

Equilibria. Unconstrained: workers solve $U'(wh)w = v'(h)$, firms $F'(L) = w$, market-clearing pins (w^*, h^*, L^*) . Constrained: cap $\bar{h} < h^*$ binds, $L^{\text{cap}} = N\bar{h}$, firm FOC gives $w^{\text{cap}} = F'(N\bar{h}) > w^*$. Define $\Delta w = w^{\text{cap}} - w^* > 0$.

Worker first-order welfare change (envelope). Using worker pre-cap envelope $\partial V_i / \partial h_i|_{h^*} = 0$: $dV_i = U'(w^*h^*)h^*\Delta w + [U'(w^*h^*)w^* - v'(h^*)]dh = U'(c_i^w)h^*\Delta w > 0$.

Firm first-order welfare change (envelope). $d\pi = [F'(L) - w]dL - Ldw = -N\bar{h}\Delta w < 0$.

Second-order Harberger DWL. Total surplus $TS(h) = N \int_0^h [F'(Nh') - v'(h')]N dh'$, maximised at h^* . Taylor to second order: $\Delta TS = -\frac{1}{2}N[v''(h^*) + |F''(L^*)|N](h^* - \bar{h})^2$.

Net. Combining under SWF:

$$\Delta \mathcal{W}^R = (\omega - 1)N\bar{h}\Delta w - \frac{1}{2}(v'' + |F''|N)(h^* - \bar{h})^2. \quad \square$$

B. Proof of Proposition 2 (unions and sectoral shocks)

Two sectors $j \in \{A, B\}$, $Y_j = z_j f(L_j)$, $f'' < 0$. Mean-preserving sectoral shock. Workers with lognormal skill $\log z_i \sim \mathcal{N}(-v_z/2, v_z)$ and log preferences.

Competitive vs union benchmarks. Competitive: $w_j = z_j f'(L_j)$, $u^{\text{comp}} = 0$, $\text{Var}^{\text{comp}}(\log w) = v_z^{\text{comp}}$. Union $w_j^U = m_j \underline{w}$ rigid downward $\Rightarrow u^U > 0$, $\text{Var}^U(\log w) = v_z^U < v_z^{\text{comp}}$.

Insurance gain. Under $\log c$ on lognormal c : $\mu_{\log c} = \mathbb{E} \log w + \log h^*$, and $\mu_{\log c}^U - \mu_{\log c}^{\text{comp}} = \frac{1}{2}\Delta v_z$ via the lognormal identity holding mean wage fixed.

Allocative loss. Each unemployed worker forgoes wh^* . Per-worker loss: $\Delta u \cdot wh^*$.

Net.

$$\Delta \mathcal{W}^U = \frac{1}{2}\Delta v_z - \Delta u \cdot wh^* > 0 \iff \Delta v_z / \Delta u > 2wh^*. \quad \square$$

C. Closed-form derivation of the integrated welfare functional (eq. 9)

Substitute $h^* = (1 - \tau_p)/(\alpha + 1 - \tau_p)$ (independent of z_i) and $c_i^* = \lambda^j [wz_i h^* (1 - u^j)]^{1 - \tau_p^j}$ into (8). Take $\mathbb{E}$ over $\log z_i$:

$$\mathbb{E}[\log c_i^*] = \log \lambda^j + (1 - \tau_p^j) [\log(wh^*(1 - u^j)) - \frac{1}{2}v_z^j].$$

The $-\frac{1}{2}(1 - \tau_p^j)v_z^j$ piece is the (I) insurance term. Adding (E) externalities, (R) redistribution, (S) SWB, (M) mortality, and (P3)/(P4) institutional channels gives the boxed expression in (9).

Sobol indices, closed form. For $\Delta \mathcal{W} = \sum_j \beta_j (X_j - \mu_j)$ with $X_j \sim U[a_j, b_j]$ independent, the first-order Sobol index is

$$S_j = \frac{\beta_j^2 (b_j - a_j)^2}{\sum_k \beta_k^2 (b_k - a_k)^2}, \quad \sum_j S_j = 1,$$

and $\Delta \mathcal{W} \stackrel{d}{\approx} \mathcal{N}(\mathbb{E} \Delta \mathcal{W}, \frac{1}{12} \sum_j \beta_j^2 (b_j - a_j)^2)$ by CLT. Numerical computation in `repl.q7_ledger.py`.

D. Proof of Proposition 3 (formal/informal welfare cost)

Setup. Worker i has fixed audit/penalty cost $\xi_i \sim G(\xi)$. Formal: $(1 - \tau)w + \sigma$. Informal: ηw , no tax, no insurance, audit cost ξ_i .

Sector choice. i informal iff $\eta w > (1 - \tau)w + \sigma - \xi_i$, i.e. $\xi_i < \xi^* \equiv w[\tau - (1 - \eta)] - \sigma$. Informal share $\phi^I = G(\xi^*)$. Comparative static $\partial \phi^I / \partial \tau = g(\xi^*)w > 0$.

Welfare cost above (C). Informal worker consumes ηwh^* vs formal $(1 - \tau)wh^* + \sigma$. The consumption part is in (C); only the foregone insurance value remains:

$$\Delta \mathcal{W}_j^F - \Delta \mathcal{W}_{US}^F = -(P^j \phi_j^I - P^{US} \phi_{US}^I) \sigma / wh^*. \quad \square$$

E. Proof of Proposition 4 (employment protection)

Setup. EPL λ_j^{EPL} raises firm cost from w to $w(1 + \lambda^{EPL})$. Heterogeneous productivity $z \sim H(z)$. Firms hire only $z \geq z^*$ where $z^* f'(L) = w(1 + \lambda^{EPL})$.

Two opposing forces. (i) Composition: $\bar{z}_{|z \geq z^*} > \bar{z} \Rightarrow$ output-per-worker rises (+); (ii) Outsider participation loss: non-hired workers go from V^w to V^n (-).

Net.

$$\Delta\mathcal{W}^P \approx \underbrace{F_z(z^*) \log(\bar{z}_{|z \geq z^*} / \bar{z})}_{(+)} - \underbrace{F_z(z^*) [\log(wh^*) - \log T_n]}_{(-)}.$$

Boeri–Garibaldi (2007, *EJ* 117(521)) reduced-form misallocation estimates suggest a parametric coefficient ≈ -0.05 :

$$\boxed{\Delta\mathcal{W}_j^P - \Delta\mathcal{W}_{US}^P \approx -0.05 \cdot (\lambda_j^{EPL} - \lambda_{US}^{EPL}).} \quad \square$$